PHY101 – GENERAL PHYSICS I

GROUP 30

2021/2022 ACADEMIC YEAR, FIRST SEMESTER

WEEK 1 LECTURE 1

BLANK – ANY QUESTIONS

UNIVERSITY OF CAPE COAST
COLLEGE OF AGRICULTURAL AND NATURAL SCIENCES
SCHOOL OF PHYSICAL SCIENCES
DEPARTMENT OF PHYSICS
PHY101 – GENERAL PHYSICS I – GROUP 30
COURSE OUTLINE FOR FIRST SEMESTER, 2021/2022 ACADEMIC YEAR
FEBRUARY, 2022
Course Code and Title: PHY 101 – GENERAL PHYSICS 1

Course Description:

This course is to introduce students to some of the fundamental concepts and principles underlying Physics so as to develop the scientific problem-solving skills and logical reasoning of students. The knowledge acquired is for later application in allied programmes like Nursing, Optometry, Computer, Science, Science Education and Laboratory Technology. The main topics treated include Physical quantities, Vectors, Dynamics, Kinematics, Thermodynamics, Work, Energy and Power.

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Course Learning Objectives:

Upon the successful completion of the course, you should be able to:

- Demonstrate knowledge of the physics content described in the course outline, and specifically the physical principles that govern the physical systems addressed.
- Demonstrate the ability to apply the appropriate physical laws to obtain qualitative understanding of a problem.
- Demonstrate the ability to express a problem in mathematical form by recognizing the important physical parameters and using equations derived from physical laws.
- describe the motion of objects in up to three dimensions in terms of their position, displacement, velocity, and acceleration.
- describe how forces change an object's motion.
- calculate the motion of an object due to the application of one or more forces.
- describe and calculate various kinds of energy and use energy to calculate the motion of objects.
- describe the fluids and their general properties.
- combine multiple concepts and apply them to real-world concepts you are likely to see in a career in science, technology, medicine, computing or engineering.
- describe heat energy and kinetic theory governing gases.
- apply dimension analysis to find unknown parameters in an equation.

Teaching/Tutorial:

10 weeks X > 2 hours of lectures + tutorials = 20+ hours

Lecture Time and Venue:

Thurs, 12:30-2:30pm, LT17

Tutorial Times

1

XXX, XXX

Grading/Evaluation: The course interactions consist of 10 lectures, 2 assessment quizzes, 2 assessment assignments, several pop or unannounced quizzes, a mini-project and credits for attendance and class participation. Some of the lectures will require you to download files from /submit files to the internet. All documents are in MS WORD/PDF/.ppt format. A minimum of 2 quizzes, 2 assignments and mini project making up the continuous assessment (CA)= 40%. The end-of-first-semester examinations consist of qualitative and quantitative problems similar to those discussed in class and those assigned for homework = 60%

Modes of Delivery: Face to Face lecture room interactions and online (Google meet, zoom). Social media interactions on WhatsApp, Telegram, Facebook, etc. will also be available.

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Course length and Pace:	Academic Integrity:
This course is delivered over 10-week period with two hours of lectures each week	Students may discuss their assignments with each other and work together on them when deemed appropriate. Each individual is expected to submit their own individual assignment you are assigned in group or groups to do group work for mini projects

Instructor Information	Teaching Assistant Information
Augustine Arthur (AA) Department of Physics School of Physical Sciences College of Agricultural and Natural Sciences University of Cape Coast **Tel. No:** 0207340766 WhatsApp: 0207340766 Telegram: 0207340766 Official Email: aarthur.ucc@gmail.com Course Email: aarthur.ucc@gmail.com	Miss Genevieve M. Nartey Tel No: 0204282962 / 0558454331 WhatsApp: 0204282962 Telegram: 0204282962 Official Email: genenart66@gmail.com

Course Materials/Reading Lists:

[HW13]. Halliday, D., Resnick, R., & Walker, J. (2013). *Fundamentals of physics*. John Wiley & Sons.

[SJ18]. Serway, R. A., & Jewett, J. W. (2018). *Physics for scientists and engineers*. Cengage learning.

[CJ14]. Cutnell, J. D., & Johnson, K. W. (2014). *Physics, Volume One: Chapters 1-17* (Vol. 1). John Wiley & Sons.

[YF13]. Young, H. D., & Freedman, R. A. (2013). Sears and Zemansky; University Physics with Modern Physics Technology.

[MS11]. S. Y. Mensah et al. (2011). *Introduction to General Physics. Volume I*. Ghana Universities Press. Legon, Accra.

General feedback: Student input is very much welcomed.

Tentative Lecture Plan

WEEK	TOPICS / DESCRIPTIONS	REMARKS
1 **27/02/2022-** **05/03/2022**	**General introduction to the course.** **Intro. to the course outline.** **Physical Quantities** ➢ Physical Quantities and Units ➢ SI units ➢ Fundamental and Derived Quantities, Standard Prefixes, Unit Conversions ➢ Dimensional Analysis ➢ Precision and Accuracy ➢ Significant Figures ➢ Rounding off ➢ Equations and Greek symbols	
2 **06/03/2022-** **12/03/2022**	**Scalars and Vectors** ➢ Types of physical quantities ➢ Components of a vector ➢ Unit vectors ➢ Position vectors ➢ Addition and subtraction of vectors ➢ Graphical addition and Subtraction of vectors ➢ Multiplication of vectors	Assignment 1

3 **13/03/2022-** **19/03/2022**	**Dynamics** ➢ Force and types of forces ➢ Newton's laws of motion ➢ Impulse and momentum ➢ Inelastic and elastic collisions ➢ Strings, pulleys and inclined planes	
4 **20/03/2022-** **26/03/2022**	**Kinematics** ➢ Displacement, speed, velocity and acceleration ➢ Graphs of motion ➢ Circular motion ➢ Simple harmonic motion ➢ Projectile motion	Quiz 1
5 **27/03/2022-** **02/04/2022**	**Work, Energy and Power** ➢ Introduction ➢ Work ➢ Energy ➢ Power	

6 **03/04/2022-** **09/04/2022**	**Elasticity** ➢ Stress and string ➢ Young's modulus ➢ Shear modulus ➢ Bulk modulus ➢ Hooke's law	Assignment 2
7 **10/04/2022-** **16/04/2022**	**Fluids** ➢ State of matter (liquid and gas) ➢ Density and Relative density ➢ Pressure in liquids ➢ Pascal's law ➢ Buoyancy and Archimedes principle	
8 **17/04/2022-** **23/04/2022**	**Heat energy** ➢ Kinetic theory of gases ➢ Temperature scales ➢ Lab and clinical thermometers compared ➢ Heat capacities of matter ➢ Thermal conductivity	Quiz 2

9 **24/04/2022-** **30/04/2022**	**Recap and Revision** ➢ Physical Quantities ➢ Scalars and Vectors ➢ Mechanics ➢ Work and Energy	
10 **01/05/2022-** **07/05/2022**	**Recap and Revision** ➢ Properties of matter ➢ Elasticity ➢ Fluids ➢ Heat	Display of CA
11 **07/05/2022-** **11/05/2022**	**Revision Week**	

12 11/05/2022 - 14/05/2022	End-of-First-Semester Examinations	
13 15/05/2022- 21/05/2022	End-of-First-Semester Examinations	
14 22/05/2022- 27/05/2022	End-of-First-Semester Examinations	
28/05/2022	Students go home	

Hints for best performance:

- Use social media channels liberally to ask questions and get answers.
- Be active on the WhatsApp and Telegram groups.
- Complete each reading assignment before the lecture.
- Use all of the resources available for the course (e.g., textbook, study guides, relevant websites)
- Don't miss class; get notes from someone if you have an unavoidable absence.
- Be an active participant in class.
- Bring your calculator every day. Use it at every opportunity.
- Complete as many exercises as you can. You learn by doing. The secret is practice.
- Before trying to solve the problems in the homework, go over my notes a couple of times and over the examples in the book.
- Before you can solve a problem, you need to understand it. Then, read the problem carefully before you try to solve it.
- Sketch the system. This visual tool will help you to understand what is going on.
- Develop a strategy for solving a given problem. Find out what concepts and principles are involved in the problem. Then develop a plan to solve the problem.
- Check results to make sure they are reasonable.
- Get help from friends, colleagues, mates or better still, come to my office, visit the physics practical lab for more help in case of physics practical—but only after trying the problem yourself.